WARING–GOLDBACH PROBLEM: ONE SQUARE, FOUR FIFTH POWERS AND TWENTY-TWO SEVENTH POWERS

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ABSTRACT. We prove that for every sufficiently large odd integer n, the equation

$$n = p_1^2 + p_2^5 + p_3^5 + p_4^5 + p_5^5 + p_6^7 + \cdots + p_{27}^7$$

has a solution with all p_j prime.

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1. INTRODUCTION

Let n be a sufficiently large positive integer that satisfies some necessary congruence conditions. Using the Hardy–Littlewood circle method, many mathematicians studied the number of representations of n as the sum of mixed powers of prime numbers. Examples of recent works on this topic include [1], [2], [3], [4], [5] and [11]. In this paper, we prove the following result.

Theorem 1.1. *For every sufficiently large odd integer n, the equation*

$$n = p_1^2 + p_2^5 + p_3^5 + p_4^5 + p_5^5 + p_6^7 + \cdots + p_{27}^7 \tag{1}$$

has a solution with all p_j prime.

2. PRELIMINARIES

Let n be a sufficiently large positive integer, $P_k = n^{\frac{1}{k}}$ and $L = \log n$. Let

$$\lambda_j = \left(\frac{193}{224}\right)^{j-1} \quad \text{for} \quad 1 \leqslant j \leqslant 12, \qquad \lambda_{13} = \frac{80375079972882501294539533063620225195 21}{480103237301210986199673678613702844088 32},$$

$$\lambda_{14} = \frac{21667849850946552228396126204623559884 7}{15003226165662843318739802456678213877 76}, \quad \lambda_{15} = \frac{58486659569630419584822001153203428 01}{46885081767696385371061882677119418368},$$

$$\lambda_{16} = \frac{790525670577515890894250054006283 81}{73257940262025602142284191682999091 2}, \quad \lambda_{17} = \frac{3427359818184343575973060164016978 5}{36628970131012801071142095841499545 6},$$

$$\lambda_{18} = \frac{59695167605853906780155489012271975}{732579402620256021422841916829990912}, \quad \lambda_{19} = \frac{26139610423872867716681836920745725}{366289701310128010711420958414995456},$$

$$\lambda_{20} = \frac{2319211726035011111954857341039457 5}{366289701310128010711420958414995456}, \quad \lambda_{21} = \lambda_{22} = \frac{5352027060080794873741978479321825}{91572425327532002677855239603748864},$$

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and write

$$\Lambda = \sum_{j=1}^{22} \lambda_j = \frac{10735022333353534120309774318126527527533695}{1536330359363875155838955771563849101082624},$$

$$\Theta = \frac{3}{10} + \frac{\Lambda}{7} = \frac{69806580440088359737857907192053053199036027}{53771562577735630454363452004734718537891840} \approx 1.29821.$$

We define

$$f_k(\alpha) = \sum_{\frac{1}{2}P_k < p \leqslant P_k} e(\alpha p^k) \log p$$

and, for $1 \leqslant j \leqslant 22$,

$$f_{k,j}(\alpha) = \sum_{\frac{1}{2}P_k^{\lambda_j} < p \leqslant P_k^{\lambda_j}} e(\alpha p^k) \log p.$$

We also define

$$F(\alpha) = f_2(\alpha) f_5(\alpha)^4 \prod_{j=1}^{22} f_{7,j}(\alpha).$$

Then by the orthogonality of additive characters, the integral

$$\int_0^1 F(\alpha) e(-\alpha n) d\alpha \tag{2}$$

counts the number of solutions of (1) with prime variables lying in certain ranges weighted by a log factor. Our goal is to show the lower bound

$$\int_0^1 F(\alpha) e(-\alpha n) d\alpha \gg n^{\Theta} \tag{3}$$

holds for all odd n.

Fix $B \geqslant 1$. Put

$$\mathfrak{M}(q,a) = \left\{ \alpha \in [0,\ 1] : \left| \alpha - \frac{a}{q} \right| \leqslant \frac{L^B}{n} \right\}, \qquad \mathfrak{M} = \bigcup_{1 \leqslant q \leqslant L^B} \bigcup_{\substack{0 \leqslant a \leqslant q \\ (a,q)=1}} \mathfrak{M}(q,a), \qquad \mathfrak{m} = [0,\ 1] \backslash \mathfrak{M}.$$

Then (3) can be obtained if we get

$$\int_{\mathfrak{M}} F(\alpha) e(-\alpha n) d\alpha \gg n^{\Theta} \tag{4}$$

and

$$\int_{\mathfrak{m}} F(\alpha) e(-\alpha n) d\alpha \ll n^{\Theta} L^{-1}. \tag{5}$$

3. The major arcs

Define

$$S_k(q,a) = \sum_{\substack{x=1 \\ (x,q)=1}}^{q} e\left(\frac{ax^k}{q} \right), \qquad v_{k,r}(\beta) = \frac{1}{k} \sum_{2^{-k} n^r < m \leqslant n^r} m^{1/k-1} e(\beta m), \qquad v_k(\beta) = v_{k,1}(\beta),$$

$$U(q,a) = S_2(q,a) S_5(q,a)^4 S_7(q,a)^{22}, \qquad w(\beta) = v_2(\beta) v_5(\beta)^4 \prod_{j=1}^{22} v_{7,\lambda_j}(\beta).$$

Lemma 3.1. *Let q, a be integers such as $1 \leqslant q \leqslant L^B$, $0 \leqslant a \leqslant q$ and $(a,q) = 1$. Then for $\alpha \in \mathfrak{M}(q,a)$, we have*

$$\sum_{\frac{1}{2}P_k^r < p \leqslant P_k^r} e(\alpha p^k) \log p = \frac{1}{\varphi(q)} S_k(q,a) v_{k,r}(\alpha - a/q) + O\left(P_k^r \exp(-c\sqrt{L}) \right)$$

for some constant $c > 0$.

Proof. This lemma follows from [[6], Lemma 6] and partial summation. □

We shall use this lemma to give an approximation of $F(\alpha)$ using functions $S_k(q,a)$ and $v_{k,r}$ defined above. By Lemma 3.1, we have

$$F(\alpha) = \frac{1}{\varphi(q)^{27}} U(q,a) w(\alpha - a/q) + O\left(n^{1+\Theta} L^{-4B}\right). \tag{6}$$

An integration over $\mathfrak{M}$ gives

$$\int_{\mathfrak{M}} F(\alpha) e(-\alpha n) d\alpha = \sum_{q \leqslant L^B} \frac{A_n(q)}{\varphi(q)^{27}} \int_{-L^B/n}^{L^B/n} w(\beta) e(-\beta n) d\beta + O\left(n^{\Theta} L^{-B}\right), \tag{7}$$

where

$$A_n(q) = \sum_{\substack{a=1 \\ (a,q)=1}}^{q} U(q,a) e(-an/q).$$

We first show that

$$\int_{-L^B/n}^{L^B/n} w(\beta) e(-\beta n) d\beta \gg n^{\Theta}. \tag{8}$$

Lemma 3.2. *([10], Lemma 6.2]). Suppose that $|\beta| \leqslant \frac{1}{2}$. Then,*

$$v_k(\beta) \ll n^{\frac{1}{k}} (1 + n|\beta|)^{-1}.$$

By Lemma 3.2 we know that

$$w(\beta) \ll n^{1+\Theta} (1 + n|\beta|)^{-2}. \tag{9}$$

By (9), we get

$$\begin{aligned}
&\int_{-L^B/n}^{L^B/n} w(\beta) e(-\beta n) d\beta \\
&= \int_{-1/2}^{1/2} w(\beta) e(-\beta n) d\beta + O\left(\int_{L^B/n}^{1/2} |w(\beta)| d\beta\right) \\
&= \int_{-1/2}^{1/2} w(\beta) e(-\beta n) d\beta + O\left(n^{\Theta} L^{-B}\right) \\
&= \frac{1}{2 \cdot 5^4 \cdot 7^{22}} \sum_{\substack{m_1 + \cdots + m_{27} = n \\ 2^{-2} n < m_1 \leqslant n \\ 2^{-5} n < m_2, m_3, m_4, m_5 \leqslant n \\ 2^{-7} n^{\lambda_{j-5}} < m_j \leqslant n^{\lambda_{j-5}} (6 \leqslant j \leqslant 27)}} m_1^{-\frac{1}{2}} (m_2 m_3 m_4 m_5)^{-\frac{4}{5}} \prod_{j=6}^{27} m_j^{-\frac{6}{7}} + O\left(n^{\Theta} L^{-B}\right) \\
&\gg \sum_{\substack{m_1 + \cdots + m_{27} = n \\ \frac{1}{4} n < m_1 \leqslant \frac{1}{2} n \\ \frac{1}{32} n < m_2, m_3, m_4 \leqslant \frac{3}{64} n \\ \frac{1}{128} n^{\lambda_{j-5}} < m_j \leqslant \frac{3}{256} n^{\lambda_{j-5}} (6 \leqslant j \leqslant 27)}} m_1^{-\frac{1}{2}} (m_2 m_3 m_4 m_5)^{-\frac{4}{5}} \prod_{j=6}^{27} m_j^{-\frac{6}{7}} \quad \gg n^{\Theta}.
\end{aligned} \tag{10}$$

Next, we want to deal with the sum

$$\sum_{q \leqslant L^B} \frac{A_n(q)}{\varphi(q)^{27}}. \tag{11}$$

Define the series

$$\mathfrak{S}(n) = \sum_{q=1}^{\infty} \frac{A_n(q)}{\varphi(q)^{27}}.$$

Lemma 3.3. *([6], Lemma 5]). For $k \geqslant 2$, we have*

$$S_k(q,a) \ll q^{\frac{1}{2}+\varepsilon}.$$

Lemma 3.4. *([10], Lemma 4.3]). For $(a,p)=1$, we have*
$$|S_k(p,a)| \leqslant ((k,p-1)-1)\sqrt{p}+1.$$

Lemma 3.5. *([6], Lemma 4]). Let $p^n \mid k$, $p^{n+1} \nmid k$ and*
$$l=\begin{cases} n+2, & p=2,\\ n+1, & p\neq 2.\end{cases}$$
If $t>l$, then
$$S_k(p^t,a)=0.$$

By Lemma 3.3, we know that $U(q,a) \ll q^{\frac{27}{2}+\varepsilon}$ for $(a,q)=1$, and hence $A_n(q) \ll q^{\frac{29}{2}+\varepsilon}$ uniformly in n. Thus, $\mathfrak{S}(n)$ converges absolutely, and
$$\sum_{q\leqslant L^B} \frac{A_n(q)}{\varphi(q)^{27}} = \mathfrak{S}(n) + O\left(L^{-B}\right). \tag{12}$$
Since the complete exponential sums are multiplicative, for $(q_1,q_2)=1$ we have
$$A_n(q_1q_2)=A_n(q_1)A_n(q_2).$$
Moreover, By Lemma 3.5 and the fact that $A_n(4)=S_5(4,a)=0$, we have $A_n(q)=0$ unless q is square-free. Hence
$$\mathfrak{S}(n)=\prod_p \mathfrak{S}_p(n)=\prod_p\left(1+\frac{A_n(p)}{(p-1)^{27}}\right). \tag{13}$$
By the orthogonality of additive characters modulo p, we have
$$\mathfrak{S}_p(n)=\frac{pM_n(p)}{(p-1)^{27}}, \tag{14}$$
where $M_n(p)$ denote the number of solutions $(x_1,\dots,x_{27})\in\left(\mathbb{F}_p^\times\right)^{27}$ of
$$x_1^2+x_2^5+x_3^5+x_4^5+x_5^5+\sum_{j=6}^{27}x_j^7 \equiv n(\mathrm{mod} p). \tag{15}$$

When $p=2$, every unit is 1, and
$$x_1^2+x_2^5+x_3^5+x_4^5+x_5^5+\sum_{j=6}^{27}x_j^7 \equiv 1(\mathrm{mod} 2).$$
Hence $n\equiv 1(\mathrm{mod}2)$ is necessary.

When $p=3$, the square is 1, and every fifth and seventh power is ±1. It is easy to see that the sum of 26 fifth and seventh powers contains all residue classes modulo 3, and (15) has a solution.

When $p=5$, we have $x^5=x$. Since $\mathbb{F}_5^\times+\mathbb{F}_5^\times=\mathbb{F}_5$, the sum of two fifth powers contains all residue classes modulo 5, and (15) has a solution.

When $p=7$, we have $x^7=x$. Since $\mathbb{F}_7^\times+\mathbb{F}_7^\times=\mathbb{F}_7$, the sum of two seventh powers contains all residue classes modulo 7, and (15) has a solution.

When $p\geqslant 11$, we define $\mathcal{A}_k=\mathcal{A}_k(p)=\{x^k : x\in\mathbb{F}_p\}$. It is known that $|\mathcal{A}_k(p)|=\frac{p-1}{(k,p-1)}$. Applying the Cauchy–Davenport theorem repeatedly, we have
$$\begin{aligned}|\mathcal{A}_2+4\mathcal{A}_5+22\mathcal{A}_7| &\geqslant \min(p,|\mathcal{A}_2|+4|\mathcal{A}_5|+22|\mathcal{A}_7|-26)\\ &\geqslant \min\left(p,\frac{p-1}{2}+\frac{4(p-1)}{5}+\frac{22(p-1)}{7}-26\right)\\ &\geqslant \min\left(p,\frac{311p-2130}{70}\right).\end{aligned}$$
Since $p\geqslant 11$, we have $\frac{311p-2130}{70}>p$, and thus $\mathcal{A}_2+4\mathcal{A}_5+22\mathcal{A}_7=\mathbb{F}_p$. Now (15) has a solution for every prime $p\geqslant 11$.

Combining above cases, we know that $M_n(p) \geqslant 1$ if n is odd, and also $\mathfrak{S}(n) > 0$ by (13). Now, by Lemma 3.4 and trivial estimates, we have

$$\begin{aligned}\left|\frac{A_n(p)}{(p-1)^{27}}\right| &\leqslant \frac{p(\sqrt{p}+1)(4\sqrt{p}+1)^4(6\sqrt{p}+1)^{22}}{(p-1)^{27}} \\ &= \frac{p^{\frac{29}{2}}}{(p-1)^{27}}\left(1+\frac{1}{\sqrt{p}}\right)\left(4+\frac{1}{\sqrt{p}}\right)^4\left(6+\frac{1}{\sqrt{p}}\right)^{22} \\ &\leqslant \frac{p^{15}}{(p/2)^{27}}\left(1+\frac{1}{\sqrt{p}}\right)\left(4+\frac{1}{\sqrt{p}}\right)^4\left(6+\frac{1}{\sqrt{p}}\right)^{22} \\ &\leqslant \frac{2^{27}\cdot 2\cdot 5^4\cdot 7^{22}}{p^{12}}\end{aligned}$$

for $p > 10^{10}$. Thus, we get

$$\prod_{p>10^{10}}\left(1+\frac{A_n(p)}{(p-1)^{27}}\right) \geqslant \prod_{p>10^{10}}\left(1-\frac{2^{28}\cdot 5^4\cdot 7^{22}}{p^{12}}\right) > c \tag{16}$$

with some $c > 0$. By (13), (14), (16) and $M_n(p) \geqslant 1$, we have

$$\mathfrak{S}(n) > c\prod_{p\leqslant 10^{10}}\frac{pM_n(p)}{(p-1)^{27}} > c\prod_{p\leqslant 10^{10}} p^{-26} \gg 1. \tag{17}$$

Now, by (7), (10), (12) and (17), the proof of (4) is completed.

4. The minor arcs

We first recall some mean value estimates proved by Andrade [3] and Kumchev [7]. Note that we can remove the logarithmic weights and the restriction to primes in $f_k(\alpha)$ and $f_{k,j}(\alpha)$ at the cost of a power of L, and deal with the classical Weyl sums.

Lemma 4.1. *We have*

$$\int_0^1 |f_2(\alpha)|^2\,|f_5(\alpha)|^6\,d\alpha \ll n^{\frac{6}{5}+\varepsilon}.$$

Proof. Taking $k_1 = k_2 = k_3 = 5$ in [[3], Lemma 4], we get Lemma 4.1. □

Lemma 4.2. *We have*

$$\int_0^1 \left|\prod_{j=1}^{22} f_{7,j}(\alpha)\right|^2 d\alpha \ll P_7^{\Lambda+\varepsilon}.$$

Proof. Taking $i = 1$ and $P = P_7$ in [[7], Lemma 1], we get Lemma 4.2. □

Write $\tau = 2^{-100}$. Define

$$\mathcal{E} = \{\alpha \in [0,\ 1] : |f_5(\alpha)| \leqslant P_5^{(3\Lambda-14)/7-10\tau}\}.$$

Using Hölder's inequality, together with Lemma 4.1, Lemma 4.2 and the definition of $\mathcal{E}$, we get

$$\begin{aligned}\int_{\mathcal{E}} |F(\alpha)|d\alpha &\leqslant \left(\int_0^1 |f_2(\alpha)|^2\,|f_5(\alpha)|^6\,d\alpha\right)^{\frac{1}{2}}\left(\int_0^1 \left|\prod_{j=1}^{22} f_{7,j}(\alpha)\right|^2 d\alpha\right)^{\frac{1}{2}}\sup_{\alpha\in\mathcal{E}}|f_5(\alpha)| \\ &\ll n^{\frac{3}{5}+\varepsilon}P_7^{\Lambda/2+\varepsilon}P_5^{(3\Lambda-14)/7-10\tau} \ll n^{\theta-\tau},\end{aligned} \tag{18}$$

where

$$\theta = \frac{139593870697983127505152898301285690218027381}{107543125155471260908726904009469437075783680} \approx 1.29803.$$

Note that $\Theta - \theta > 0.0001$.

5. Proof of Theorem 1.1

Lemma 5.1. *Suppose that $X^{\frac{1}{24}} \leqslant P \leqslant X^{\frac{9}{20}}$. Then either we have*

$$\sum_{X<p\leqslant 2X} e(\alpha p^5) \ll X^{\frac{47}{48}+\varepsilon},$$

or there exist integers q and a such that $1 \leqslant q \leqslant P$, $(a,q)=1$ and $|q\alpha - a| \leqslant PX^{-5}$.

Proof. Taking $k=5$ in [[9], Lemma 2.2], we get Lemma 5.1. □

Now suppose that $\alpha \in [0,\ 1]\backslash\mathcal{E}$. We have $|f_5(\alpha)| > P_5^{(3\Lambda-14)/7-10\tau}$ in this case. Taking $X = \frac{1}{2}P_5$ and $P = \frac{1}{32}n^{\frac{1}{119}}$ in Lemma 5.1, we find that the first case will never occur since $\frac{3\Lambda-14}{7} - \frac{47}{48} > 0.01$. Hence, we must have

$$[0,\ 1]\backslash\mathcal{E} \subset \mathfrak{R} = \bigcup_{1\leqslant q\leqslant \frac{1}{32}n^{\frac{1}{119}}} \bigcup_{\substack{0\leqslant a\leqslant q\\ (a,q)=1}} \mathfrak{R}(q,a), \tag{19}$$

where

$$\mathfrak{R}(q,a) = \{\alpha \in [0,\ 1] : |q\alpha - a| \leqslant n^{-\frac{118}{119}}\}. \tag{20}$$

Define a function $\Upsilon : \mathfrak{R} \to [0,\ 1]$ by

$$\Upsilon(\alpha) = (q + n|q\alpha - a|)^{-1}.$$

Taking $k=2,5$ in [[8], Theorem 2] and by partial summation, we have

$$f_2(\alpha) \ll P_2 L^C \Upsilon(\alpha)^{\frac{1}{2}-\varepsilon} \tag{21}$$

and

$$f_5(\alpha) \ll P_5 L^C \Upsilon(\alpha)^{\frac{1}{2}-\varepsilon} \tag{22}$$

for some constant $C>0$. By (21), (22) and trivial upper bounds for $f_{7,j}(\alpha)$ $(1 \leqslant j \leqslant 22)$, we have

$$F(\alpha) = f_2(\alpha) f_5(\alpha)^4 \prod_{j=1}^{22} f_{7,j}(\alpha) \ll n^{1+\Theta} L^{27C} \Upsilon(\alpha)^{\frac{49}{20}}. \tag{23}$$

Integrating over $\mathfrak{R}\backslash\mathfrak{M}$ gives

$$\int_{\mathfrak{R}\backslash\mathfrak{M}} |F(\alpha)| d\alpha \ll n^{1+\Theta} L^{27C} \int_{\mathfrak{R}\backslash\mathfrak{M}} |\Upsilon(\alpha)|^{\frac{49}{20}} d\alpha \ll n^{\Theta} L^{27C-\frac{9}{20}B+1}. \tag{24}$$

Choosing $B = 200C + 100$ in (24) and by the fact that $\mathfrak{m} \subset \mathcal{E} \cup \mathfrak{R}\backslash\mathfrak{M}$, the proof of (5) is completed. From (4) and (5) we get (3), and the proof of Theorem 1.1 is completed.

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